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Step 15 - Timer Expiry Logic

Reference Note

This chapter is based on concepts described in the ARM Generic Timer Architecture documentation.

Previous chapters explained:

- Physical and Virtual Counters
- Physical and Virtual Timers
- TVAL vs CVAL
- Timer Ownership

This chapter explains:

How does a timer know it has expired?
What happens internally before an interrupt is generated?
What do ENABLE, IMASK, and ISTATUS actually do?

Understanding this logic is essential because many timer debugging issues stem from misunderstanding the timer state machine.

Learning Objectives

After completing this chapter, you should understand:

- Timer compare logic
- Expiry conditions
- ENABLE bit
- IMASK bit
- ISTATUS bit
- Timer state transitions
- Why timer interrupts are level-sensitive
- Common timer debugging scenarios

1. Timer Expiry Overview

Every Generic Timer contains compare logic.

The timer continuously evaluates:

Value 1	Value 2
Current Counter	Compare Value (CVAL)

When:

Current Counter >= Compare Value

the timer expires.

2. Timer State Machine

A timer can be viewed as existing in three states.

State	Description
Disabled	Timer not operating
Running	Timer waiting for expiry
Expired	Compare condition satisfied

State Transition Flow

Current State	Event	Next State
Disabled	ENABLE=1	Running
Running	Compare Match	Expired
Expired	New TVAL/CVAL programmed	Running
Expired	ENABLE=0	Disabled

3. Compare Logic

Suppose:

Parameter	Value
Current Count	1,000,000
Compare Value	1,024,000

The timer repeatedly evaluates:

1,000,000 >= 1,024,000 ?

Answer:

No, Timer continues waiting.

As the counter advances:

1,024,000 >= 1,024,000 ?

Answer:

Yes, Timer expires.

4. CNTP_CTL_EL0 / CNTV_CTL_EL0

Both Physical and Virtual Timers contain the same control fields.

Field	Purpose
ENABLE	Enable or disable timer
IMASK	Mask interrupt output
ISTATUS	Expiry status

5. ENABLE Bit

The ENABLE bit controls whether the timer operates.

ENABLE = 1

Behavior
Timer active
Compare logic running
Expiry detection enabled

ENABLE = 0

Behavior
Timer disabled
Compare logic inactive
No timer event generation

Example

ENABLE	Result
1	Timer running
0	Timer stopped

6. IMASK Bit

IMASK controls interrupt signaling.

Important:

IMASK does NOT stop the timer.

Many engineers misunderstand this behavior.

IMASK = 0

Behavior
Interrupt output enabled

IMASK = 1

Behavior
Interrupt output masked

Important

The timer still:

Counts

Compares
Expires
Updates ISTATUS

Only interrupt signaling is suppressed.

Example

Field	Value
ENABLE	1
IMASK	1
Timer Expired	Yes

Result:

ISTATUS = 1
No interrupt delivered

7. ISTATUS Bit

ISTATUS indicates whether the timer expiry condition is true.

Think of ISTATUS as:

Has the timer expired?

ISTATUS = 0

Meaning
Timer not expired

ISTATUS = 1

Meaning
Timer expired

Example

Current Count	Compare Value	ISTATUS
100	200	0
150	200	0
200	200	1
250	200	1

8. Timer Expiry Flow

Step	Action
1	Timer enabled
2	Counter advances
3	Compare condition checked
4	Current Count ≥ Compare Value
5	ISTATUS becomes 1
6	Interrupt output asserted (if IMASK = 0)

9. Why Interrupts Are Level-Sensitive

One of the most important concepts in Generic Timer behavior.

After expiry:

Current Count >= Compare Value

remains true.

Example:

Current Count	Compare Value
1,024,000	1,024,000
1,024,100	1,024,000
1,024,200	1,024,000

The condition is still satisfied.

Therefore:

Timer continues signaling

until software takes action.

10. How To Clear An Expired Timer

The timer clears naturally when software:

Option 1

Programs a new deadline.

Example:

CNTP_CVAL = New Value

or

CNTP_TVAL = New Interval

The compare condition may become false again.

Option 2

Masks interrupt output.

IMASK = 1

Option 3

Disables timer.

ENABLE = 0

11. Practical Debugging Table

Symptom	Possible Cause
No Interrupt	ENABLE = 0
No Interrupt	IMASK = 1
No Interrupt	Compare value in future
Immediate Interrupt	Compare value already expired
Repeated Interrupts	Timer not reprogrammed
ISTATUS = 1	Timer expired

Architecture Summary

Component	Responsibility
Counter	Measures time
Compare Logic	Detects expiry
ENABLE	Activates timer
IMASK	Masks interrupt output
ISTATUS	Indicates expiry
Timer Expiry	Current Count ≥ Compare Value

Common Misconceptions

Misconception 1

IMASK stops the timer

False.

It only masks interrupt output.

Misconception 2

ISTATUS means interrupt delivered

False.

ISTATUS only means:

Timer expired

Interrupt delivery involves additional hardware.

Misconception 3

Timer expiry automatically clears itself

False.

Software usually must program a new deadline.

Misconception 4

Timer interrupt is edge-triggered

False.

Generic Timer behavior is level-sensitive.

Key Takeaways

Concept	Key Point
Expiry Condition	Current Count ≥ Compare Value
ENABLE	Starts timer
IMASK	Masks interrupt output
ISTATUS	Indicates expiry
Timer Continues Running	Even when IMASK=1
Generic Timer Interrupt	Level-sensitive

Knowledge Check

Q1

What condition causes timer expiry?

Q2

What does ENABLE control?

Q3

Does IMASK stop timer counting?

Q4

What does ISTATUS indicate?

Q5

Why can timer interrupts repeatedly occur after expiry?

Answer Key

Question	Answer
Q1	Current Count ≥ Compare Value
Q2	Whether the timer is active
Q3	No
Q4	Whether the timer has expired
Q5	The compare condition remains true after expiry